

Francisco Patitucci

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Education

University of Texas at Austin, PhD in Electrical Engineering Fall 2023 – Spring 2028
Optimization for Machine Learning. Advisor: **Prof. Aryan Mokhtari**. (Expected)
• Selected coursework: Convex Optimization, Advanced Probability, Statistical Machine Learning.
Universidad de Buenos Aires, MSc in Mathematical Engineering, Signal Processing Fall 2021 – Transferred to UT Track
• Selected coursework: Stochastic Processes, Statistical Learning, Optimization, Matrix Analysis.
Universidad Nacional de La Plata, Engineering Degree in Aerospace Spring 2016 – 2021

Experience

Graduate Research Assistant — Aryan Mokhtari Group, UT Austin Fall 2023 – Present
• Developed theoretical and empirical tools for stochastic, nonconvex, and adaptive optimization methods.
• Designed and implemented adaptive optimization algorithms in Python/PyTorch for deep learning.
• Ran large-scale optimizer benchmarks on HPC clusters using Slurm, evaluating convergence, stability, and robustness on language models up to 350M parameters.
• Built reproducible experiment pipelines for learning-rate sweeps, momentum configurations, random seeds, and model-scale comparisons.
Research Assistant — Paul Puleston Group, UNLP Jan 2021 – Aug 2022
• Implemented and deployed a turbine-emulator control module in C/MATLAB, improved the user interface, and deployed the system to a microcontroller.

Selected Publications

F. Patitucci, A. Mokhtari. *Adaptive Optimization via Momentum on Variance-Normalized Gradients*. arXiv preprint, 2026.
F. Patitucci, R. Jiang, A. Mokhtari. *Improving Online-to-Nonconvex Conversion for Smooth Optimization via Double Optimism*. **ICLR 2026**.
R. Jiang, A. Mokhtari, **F. Patitucci**. *Improved Complexity for Smooth Nonconvex Optimization: A Two-Level Online Learning Approach with Quasi-Newton Methods*. **STOC 2025**.

Honors and Awards

Gold Reviewer Award — International Conference on Machine Learning (ICML) 2026
Graduate Research Fellowship — UT Austin Fall 2023 – Fall 2027
Travel Award — ACM Symposium on Theory of Computing (STOC) 2025
Fulbright Scholarship (declined) — Fulbright Argentina 2023
Ing. Isidoro Marín Medal — National Academy of Engineering, Argentina 2022

Skills

Programming: Python, PyTorch, NumPy, MATLAB, C/C++ basics
ML / Systems: optimizer implementation, distributed training, language-model training, hyperparameter sweeps
Infrastructure: Slurm/HPC, Linux, Git, Bash, $\LaTeX$
Languages: Spanish (native), English (fluent)